IntelQCopter

Autonomous Quadcopter

TECHNICAL DESIGN AND VALIDATION REPORT



AI CONCEPT ILLUSTRATION / RED AND WHITE F450 / NOT A HARDWARE TEST PHOTO

F450	1 kHz	ESP32-S3	ROS 2
X FRAME	FC TARGET	CURRENT FC	OPTIONAL

Visual models, calibration, estimation and perception-to-control architecture.

Equations, compact graphs, materials and traceable engineering decisions.

Current status: sensor and software groundwork; flight acceptance pending.

IQC-PPP-002 / 28 OCTOBER 2026 / UNIVERSITY SUBMISSION DRAFT

Student / ID / institution / supervisor / academic year: complete before submission.

Proposed targets and historical results are distinguished. Motor outputs remain disabled.

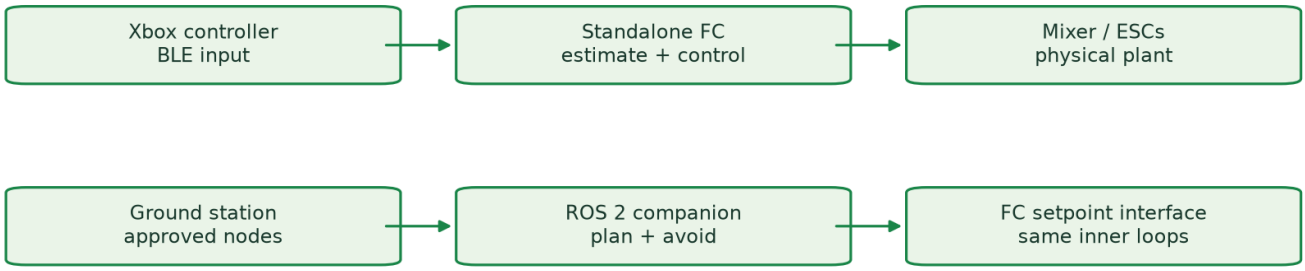
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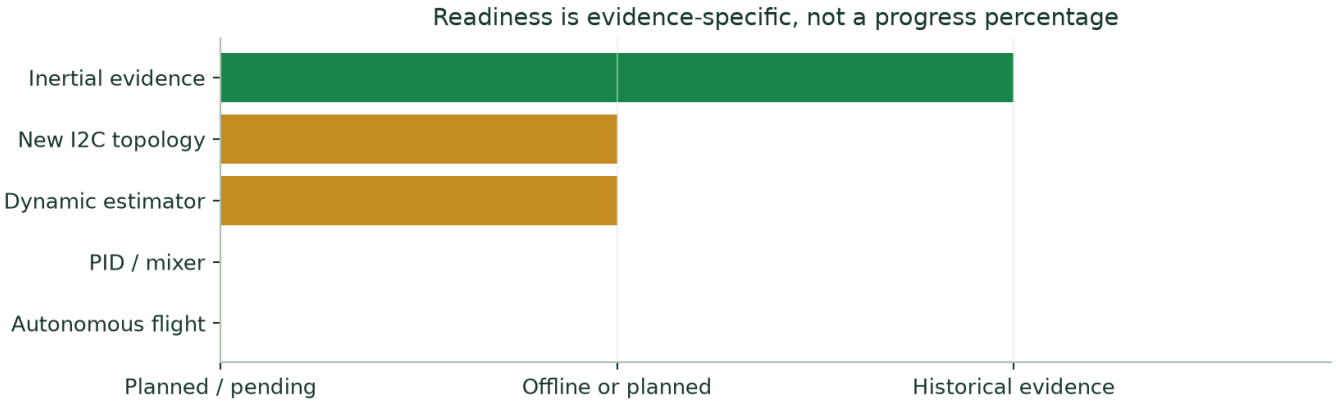
Figure label	Interpretation
Real log / historical fit	Derived from recorded project evidence; reference limits remain
Existing synthetic model	A prior numerical experiment, not flight or Gazebo
Analytic example / schematic	Explains a law or geometry under stated assumptions
Proposed architecture / campaign	Intended design or test, not a completed capability

Project in one page

Objective: a standalone stable-flight controller, extended progressively to autonomous waypoint navigation and obstacle avoidance. The aircraft must move through the actual FC command path.



PROPOSED ARCHITECTURE — Two command paths; motor commands belong only to the FC. The companion is optional.

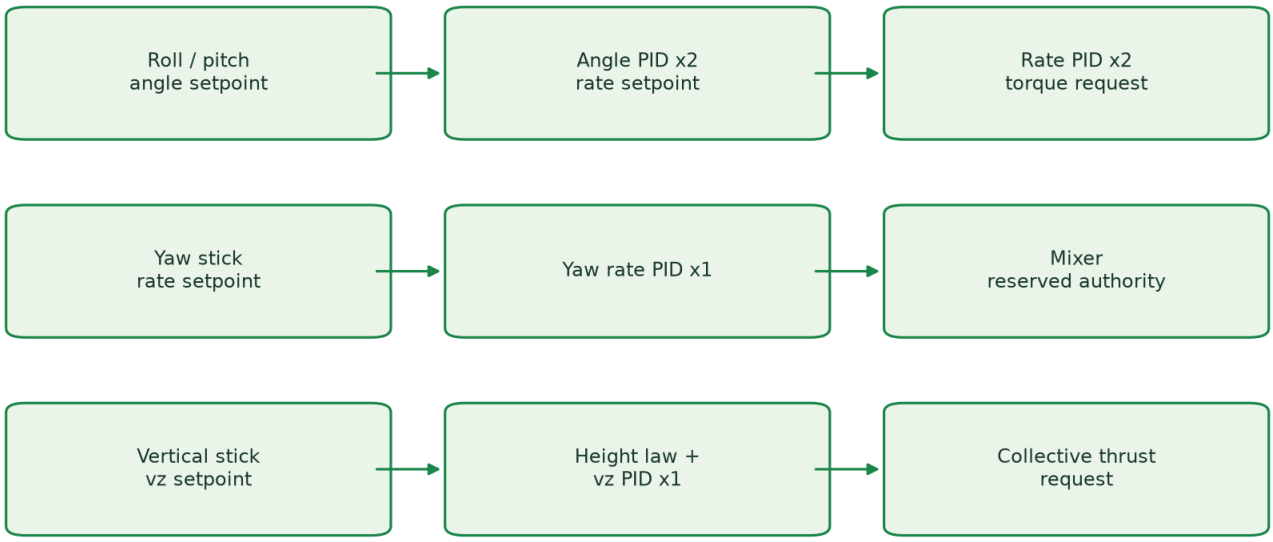


PROJECT STATUS SUMMARY — Historical evidence does not imply flight acceptance. New sensor wiring and physical frame checks remain pending.

D = user decision; **P** = proposed target; **TBD** = unresolved. Historical hardware evidence, offline replay, synthetic models and planned architectures are labelled separately. No new hardware or flight experiment was run for this edition.

Reading map: physics -> calibration -> estimation -> control -> perception/navigation -> verification. Complete requirements and materials follow the visual chapters. Pin wiring and low-level syntax are outside the main academic narrative; the preserved engineering baseline remains available as a reference.

Flight behaviour and control architecture



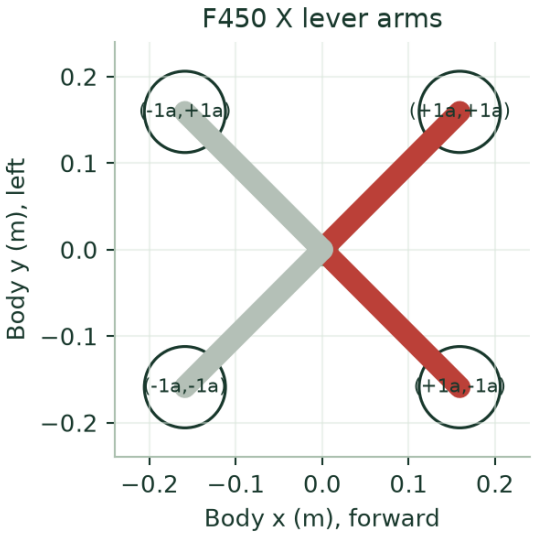
PROPOSED ARCHITECTURE — Six full PID blocks are requested. The outer height-to-speed law remains a design choice.

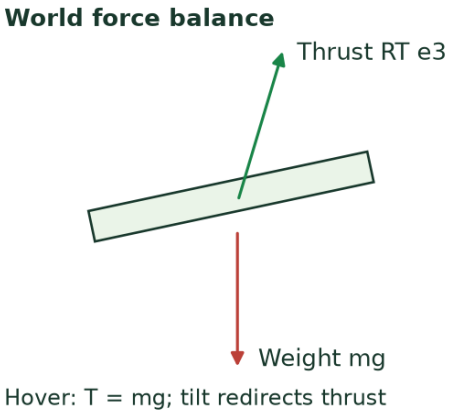
$$e_k = r_k - y_k, \qquad u_k = K_p e_k + K_i \sum_{j=0}^{k} e_j \Delta t_j + K_d \dot{e}_{f,k}$$

Pilot input	Controlled variable	Neutral behaviour
Right stick	Roll and pitch angles	Level attitude; no horizontal brake or position hold
Left horizontal	Yaw rate	Zero commanded yaw rate; no absolute-heading hold
Left vertical	Vertical speed	Height retention through the vertical law
RT / LT	Bounded increase / decrease	Gentle profile; reserve correction authority

Six PID blocks: two angle, two rate, one yaw rate and one vertical-speed PID. An outer proportional height-to-speed law is an option, not a frozen implementation. Mode transitions, boost interaction and vertical saturation still require design review.

F450 geometry and force balance





ANALYTIC GEOMETRY — Frame convention is explicit; this drawing does not assign motor numbers or spin signs.

$$L = 0.225 \text{ m}, \qquad a = \frac{L}{\sqrt{2}} \approx 0.1591 \text{ m}$$

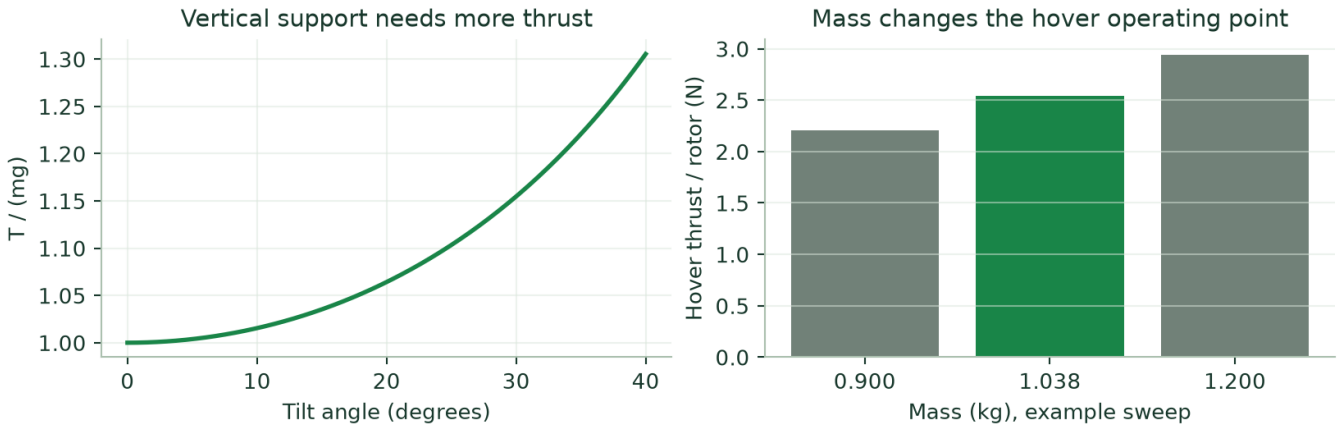
$$T = \sum_{i=1}^{4} T_i, \quad m\dot{\mathbf{v}} = R\mathbf{e}_3 T - mg\mathbf{e}_3 - D\mathbf{v} + \mathbf{f}_{ext}$$

$$J\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (J\boldsymbol{\omega}), \qquad \boldsymbol{\tau}_i = \mathbf{r}_i \times \mathbf{F}_i$$

Meaning: sum the rotor forces in one reference frame; subtract gravity; the remaining force accelerates the mass. Lever arms convert each rotor force into roll/pitch torque. Rotor reaction torque contributes yaw. The cross term is angular-momentum coupling in rotating body axes.

Frame proposal: ENU world, FLU body. Confirm calibration conventions and physical motor corners before mixing. Geometry is nominal; CW/CCW signs are not assigned by this illustration.

Hover operating point and uncertain parameters



ANALYTIC EXAMPLE — At the supplied mass, hover requires about 2.545 N per rotor; actual capability is unmeasured.

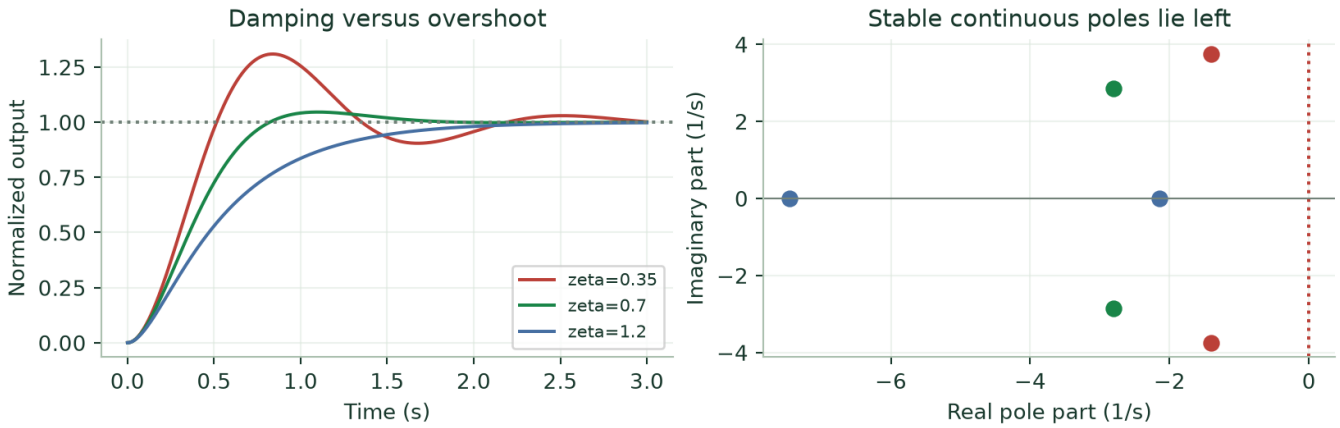
$$T_0 = mg \approx 10.179 \text{ N}, \qquad T_{i,0} \approx 2.545 \text{ N} \quad (m = 1.038 \text{ kg})$$

Parameter	Supplied model value	Qualification
Mass	1.038 kg; stated sigma 0.0160312 kg	Simulation input; verify mass and covariance assumptions
Jxx, Jyy	0.006421 kg m squared	Approximate inertia about centre of mass
Jzz	0.011853 kg m squared	Products of inertia not established
Battery	3S 4200 mAh; marked 120C	320 g in parameter file versus old rough 150 g estimate; weigh
Propulsion	1400 KV; 9047; 40 A ESCs	Exact motor and thrust/command response unknown

$$J_{xx} = \sum_i [J_{xx,i}^{COM} + m_i(y_i^2 + z_i^2)], \qquad \Sigma_y \approx G\Sigma_p G^T$$

Measurement gate: complete mass, centre of mass, motor positions, thrust curve, actuator lag, torque coefficient and power demand. KV, ESC amperage and C markings cannot supply those measurements. Propulsion tests require separate authorization and a suitable fixture.

Linear model and intuitive stability



ILLUSTRATIVE LTI MODEL — Educational second-order responses, not tuned drone gains or simulated aircraft trials.

$$G_\phi(s) = \frac{1}{J_{xx}s^2}, \quad G_\theta(s) = \frac{1}{J_{yy}s^2}, \quad G_r(s) = \frac{1}{J_{zz}s}, \quad G_z(s) = \frac{1}{ms^2}$$

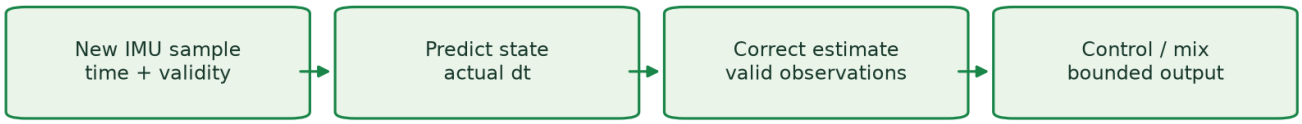
$$G_{act}(s) = \frac{1}{\tau_m s + 1}, \qquad s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$K_p = J\omega_n^2, \qquad K_d = 2\zeta\omega_n J \quad \text{(ideal torque PD example)}$$

These transfer functions are local hover approximations, not global flight models. The step-response plot illustrates damping; it does not identify drone gains. Add measured actuator lag, filtering, delays, voltage effects and saturation before design acceptance.

Proposed small-signal targets: phase margin >=45 degrees; gain margin >=6 dB; overshoot <=10 percent; settling within 5 percent <=1 s. Final six-loop numerical gains remain TBD pending identification. Compare the local model with nonlinear physics rather than assuming the two are identical.

Discrete controller and sample timing

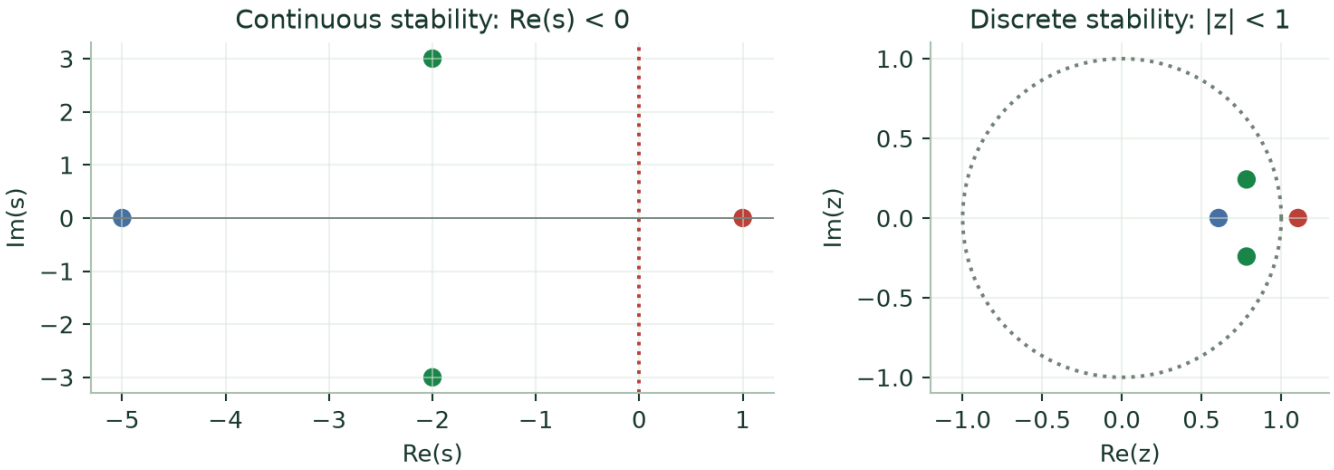


PROPOSED ARCHITECTURE — Sample, estimate, control and ESC update are distinct stages; slow observations are asynchronous.

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \implies \mathbf{x}_{k+1} = A_d\mathbf{x}_k + B_d\mathbf{u}_k$$

$$A_d = e^{A\Delta t}, \qquad B_d = \int_0^{\Delta t} e^{A\tau} B \, d\tau$$

$$\phi_{k+1} = \phi_k + \Delta t \, p_k + \frac{\Delta t^2}{2J_{xx}}\tau_{x,k}, \qquad p_{k+1} = p_k + \frac{\Delta t}{J_{xx}}\tau_{x,k}$$

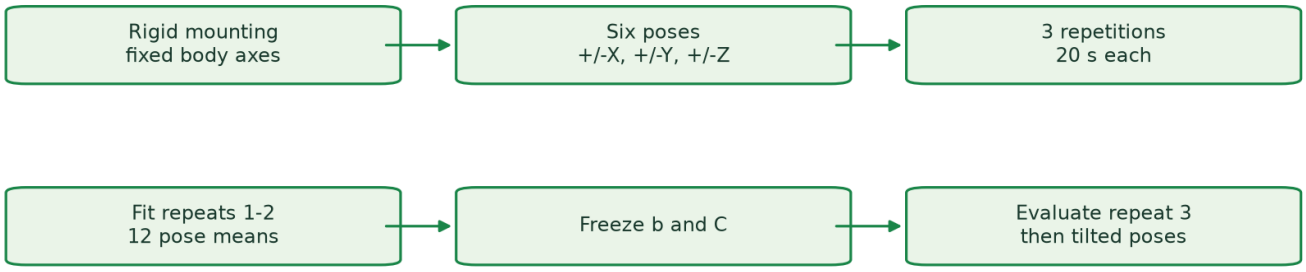


ANALYTIC SAMPLING EXAMPLE — This example uses 100 ms to show the mapping clearly; it is not the 1 ms FC implementation or its poles.

Target period: 1 ms. Timestamp acquisition, estimation, PID and output separately. Slow sensors update at their own cadence; an old sample is not new because the controller reads it again. ESC command rate depends on the actual ESC protocol, not its 40 A rating.

Implementation rule: coherent state snapshots, bounded queues/timeouts and one owner per I2C bus. A full MPU reset requires bypass restoration before accessing downstream sensors.

Calibration experimental design



HISTORICAL TEST DESIGN — Existing 18-pose design separates fitting and evaluation. Fixture uncertainty remains.

$$\mathbf{r}_j = \mathbf{b} + A\mathbf{a}_j + \boldsymbol{\varepsilon}_j, \qquad \mathbf{a}_j \in \{\pm g\mathbf{e}_x,\ \pm g\mathbf{e}_y,\ \pm g\mathbf{e}_z\}$$

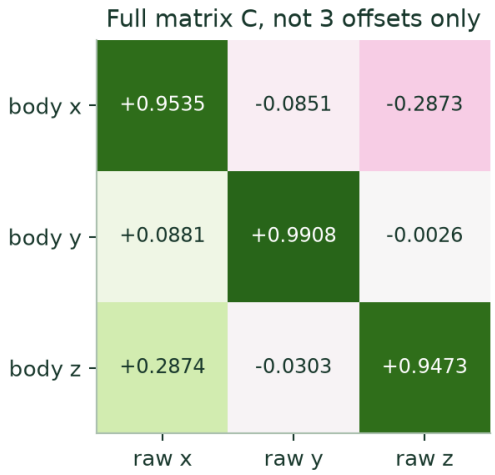
$$H = [\,\mathbf{h}_1 \cdots \mathbf{h}_{12}\,], \quad \mathbf{h}_j = [1, a_{x,j}, a_{y,j}, a_{z,j}]^T, \quad Y = [\,\mathbf{r}_1 \cdots \mathbf{r}_{12}\,]$$

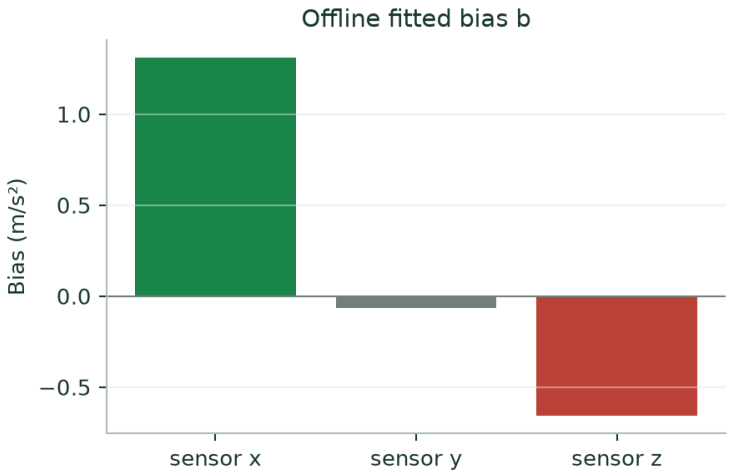
$$\widehat{M} = YH^T(HH^T)^{-1} = [\,\widehat{\mathbf{b}}\ \widehat{A}\,], \qquad C = \widehat{A}^{-1}$$

Identifiability: H must have full row rank. Six independent gravity directions reveal offset and directional response; one level pose cannot identify the full matrix. Equal fit weights do not imply exact pose references.

Evidence [P1]: 18 recordings, 20 seconds each; repetitions 1-2 fit the candidate and repetition 3 is reserved. A detached mounting invalidates its associated dataset. The secured-mount data and later tilted-pose review remain distinct. Standard gravity 9.80665 m/s squared is an assumption, not a local gravimeter reading.

Full matrix correction and held out residuals





HISTORICAL OFFLINE FIT — Coefficients shown are the historical fit candidate, not a new calibration or an instruction to overwrite trims.

$$\mathbf{a}_{cor} = C(\mathbf{r} - \mathbf{b})$$



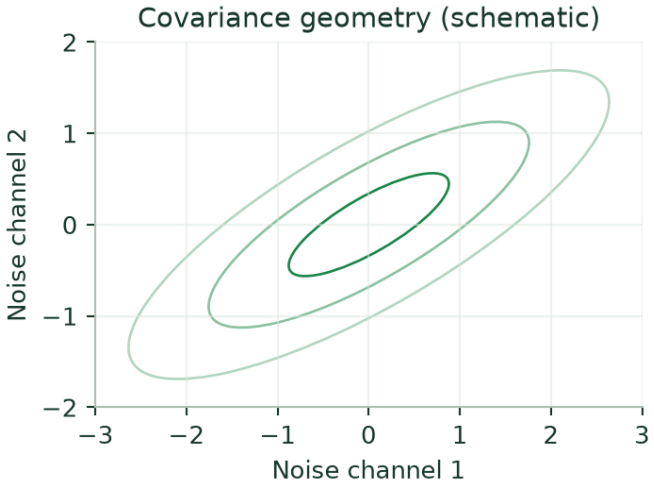


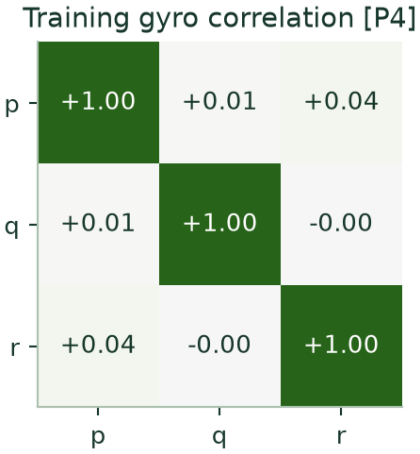
REAL LOG REPLAY SUMMARY — These residuals include reference/placement error; they are not certified sensor accuracy.

Read the heatmap by row: each corrected body axis may depend on all three raw axes. C includes scale/cross-axis/mounting effects in this historical fit. The bias and matrix are not interchangeable with three MPU offset registers.

Do not recalibrate from this page. These are historical offline-fit coefficients [P1], not a newly installed calibration. Residuals contain fixture/placement uncertainty; they are not certified sensor accuracy. Frozen firmware constants require their own recorded provenance and reset/power-cycle verification.

Covariance and useful information

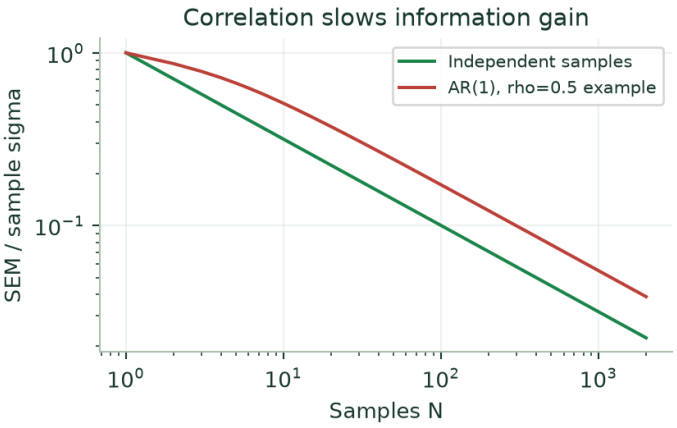


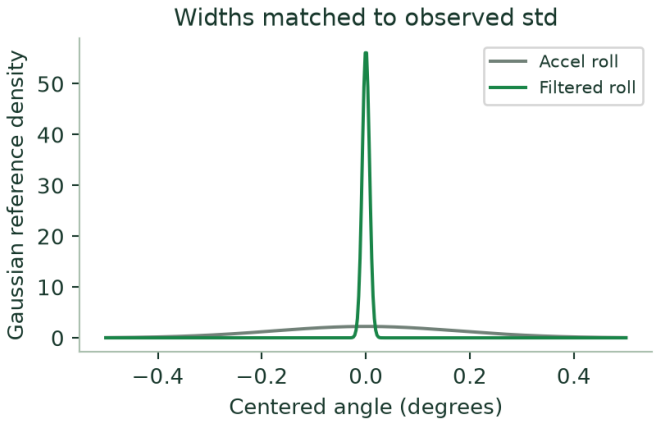


SCHEMATIC + MEASURED SUMMARY — Off-diagonal terms show shared variation. Zero covariance does not generally imply independence.

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_i \mathbf{x}_i, \qquad S = \frac{1}{N-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

$$\mathrm{Cov}(\bar{\mathbf{x}}) = \frac{1}{N^2}\sum_i \sum_j \mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j)$$

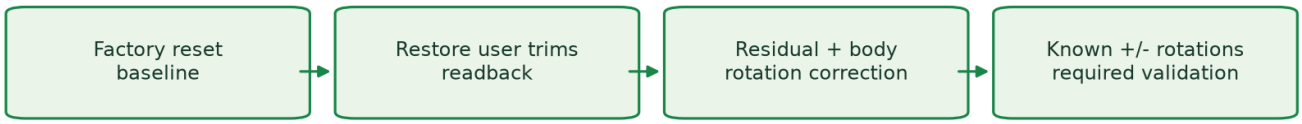




ANALYTIC EXAMPLE + OBSERVED WIDTHS — Density shapes are explanatory references, not a claim that the actual noise is Gaussian or unbiased.

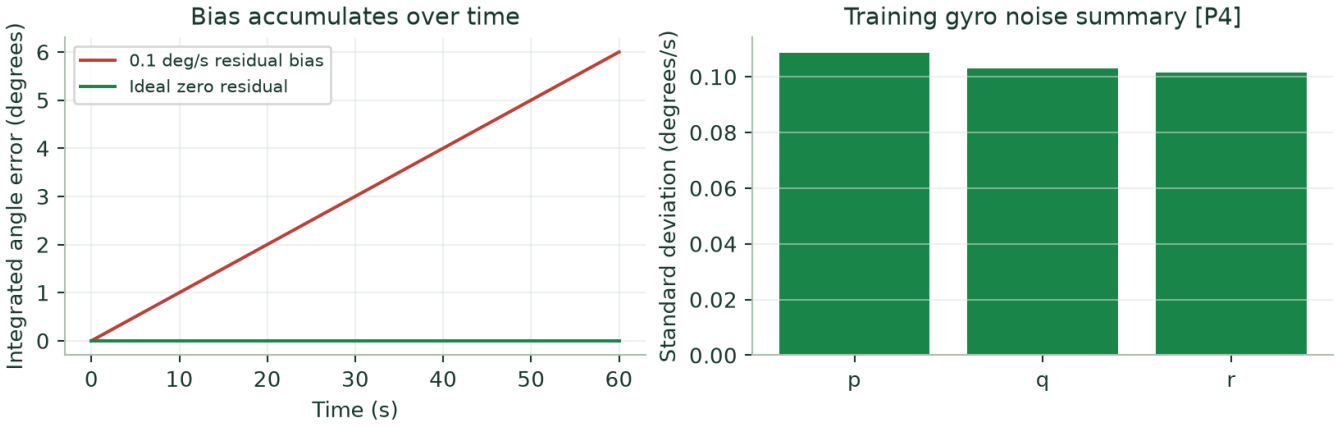
For independent samples, the last expression reduces to Sigma/N. With correlation, cross-time terms remain. More samples do not automatically remove drift, reference error or modelling bias. The density curves show widths only; neither joint Gaussianity nor independence is asserted. Report repeatability and accuracy separately.

Gyro calibration and frame validation



PROPOSED ARCHITECTURE — Hardware bias correction and software transform have separate roles; gyro scale/physical axes remain pending.

$$\boldsymbol{\omega}_b = R_{bs}(\boldsymbol{\omega}_s - \mathbf{b}_g), \qquad \Delta\theta_{bias}(t) = b_g t$$

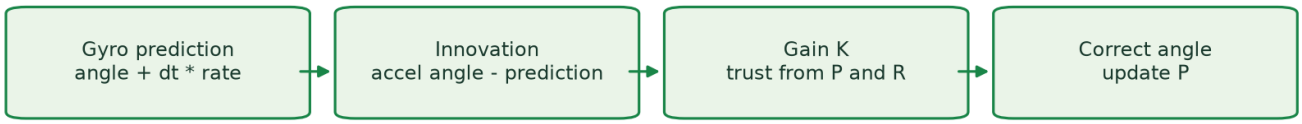


ANALYTIC EXAMPLE + MEASURED SUMMARY — Averaging reduces random fluctuation but does not remove an unmodelled residual bias.

Bias is not scale or frame mapping. Motionless data estimate bias/repeatability. Known positive and negative rotations about physical body X, Y and Z test sign and mapping; a measured rotation reference is additionally needed for scale.

MPU user offsets are volatile. Restore against the documented reset baseline, read back, then apply only the remaining software bias. Do not subtract the full bias twice or use the accelerometer scale matrix as a gyro rotation. Physical gyro direction/scale checks remain pending.

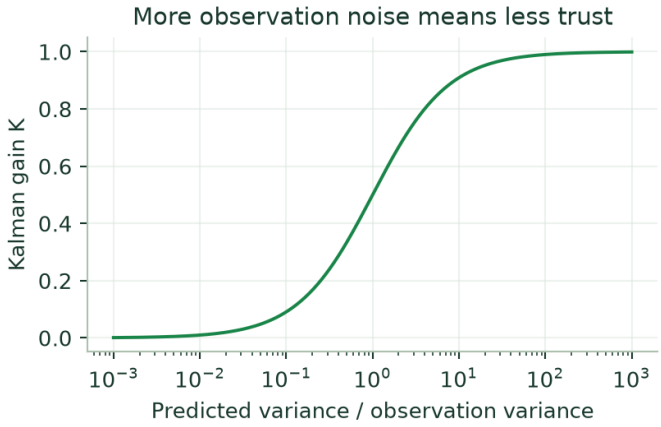
Scalar Kalman filter and trust

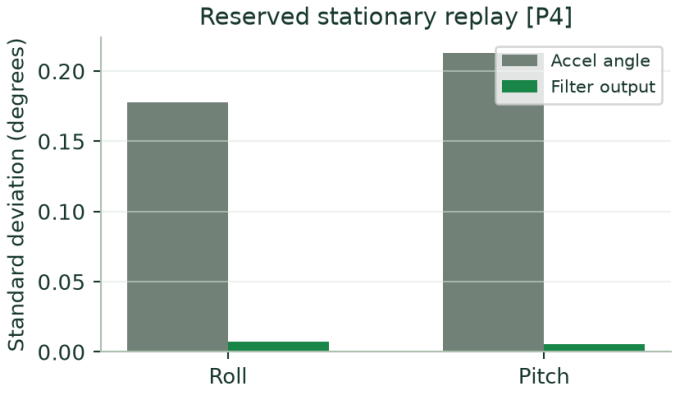


PROPOSED ARCHITECTURE — Scalar roll/pitch estimator candidate. Translational acceleration can corrupt the gravity observation.

$$\theta_k^- = \theta_{k-1} + \Delta t\,\omega_k, \quad P_k^- = P_{k-1} + Q_k, \quad \nu_k = z_k - \theta_k^-$$

$$K_k = \frac{P_k^-}{P_k^- + R_k}, \qquad \theta_k = \theta_k^- + K_k\nu_k, \qquad P_k = (1 - K_k)P_k^-$$



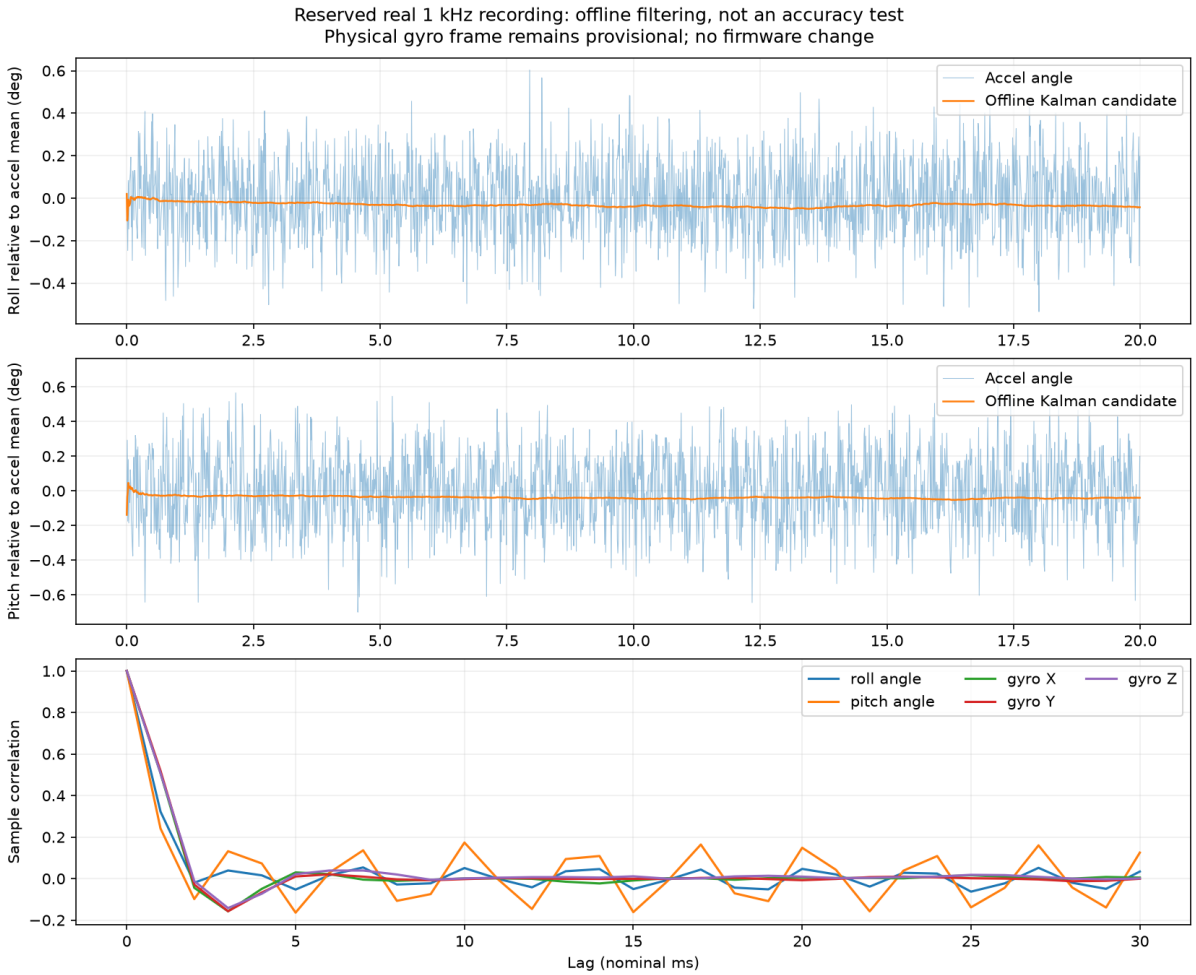


ANALYTIC LAW + REAL REPLAY SUMMARY — Smoother output is not an independent accuracy result; mean errors and correlation remain.

Historical scalar model [P4]	Q (rad squared / step)	R (rad squared)	Conditional limiting K
Roll	3.5948594e-12	9.6143606e-6	0.00061129
Pitch	3.2286779e-12	1.3818086e-5	0.00048326

For independent discrete gyro-rate noise, angle-process variance is sigma_rate squared times dt squared. Continuous white-rate spectral density instead integrates proportionally to dt. Keep those models and units distinct. Existing candidate gains are model-dependent; stationarity and autocorrelation limit their interpretation [P4].

Original recording replay

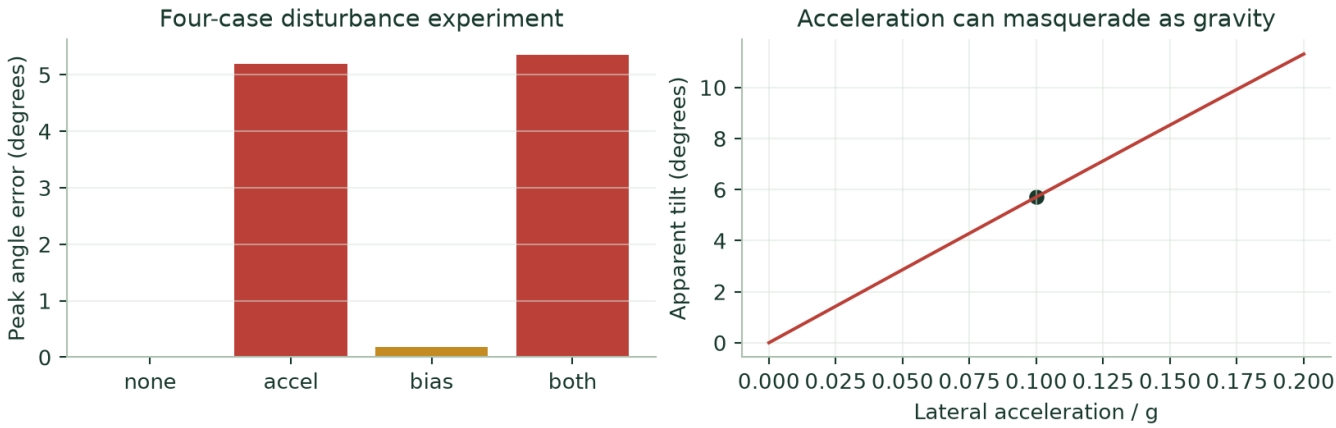


Original derived figure [P4], reproduced without alteration. Engine: Python float64 replay of a reserved real 1 kHz recording. This is not live sensor output, an MCU filter execution or a flight. All samples enter the analysis; displayed points are reduced for readability.

Reserved result	Roll (degrees)	Pitch (degrees)
Acceleration-angle standard deviation	0.177569	0.213074
Filter-output standard deviation	0.007000	0.005692
Filter mean minus observation mean	-0.033769	-0.039908

The evaluation excludes a fixed two-second initialization interval. No independently measured attitude reference exists for this recording; lower fluctuation does not demonstrate absolute accuracy. Mean differences were retained, not fitted away. The lower panel shows why independent-white-noise assumptions need checking.

Why dynamic acceleration defeats a quiet filter



EXISTING SYNTHETIC MODEL + ANALYTIC LAW — Python model results, not Gazebo or flight. The candidate is not accepted for dynamic operation.

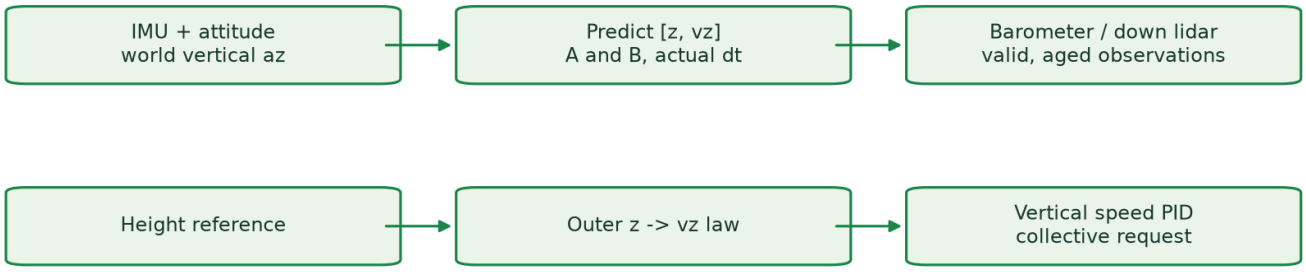
$$\theta_{apparent} = \arctan(a_{lat}/g), \qquad \|\mathbf{a}\|/g = \sqrt{1 + (a_{lat}/g)^2}$$

At a_lat = 0.1g, apparent tilt is 5.71 degrees while the norm is only about 1.005g. A magnitude-only acceptance gate can miss the disturbance.

Existing model case [P4]	RMSE (degrees)	Peak error (degrees)
No added disturbance	0.004312	0.012507
0.1g acceleration	2.114109	5.191244
0.1 degree/s gyro bias	0.157612	0.172872
Both	2.214463	5.354520

Decision: evaluate innovations, bias dynamics and acceleration-sensitive correction on independent dynamic references before acceptance. These are existing Python synthetic-model results; no Gazebo or physical flight is implied. Smoothing or collecting more stationary samples cannot remove an unobservable force component.

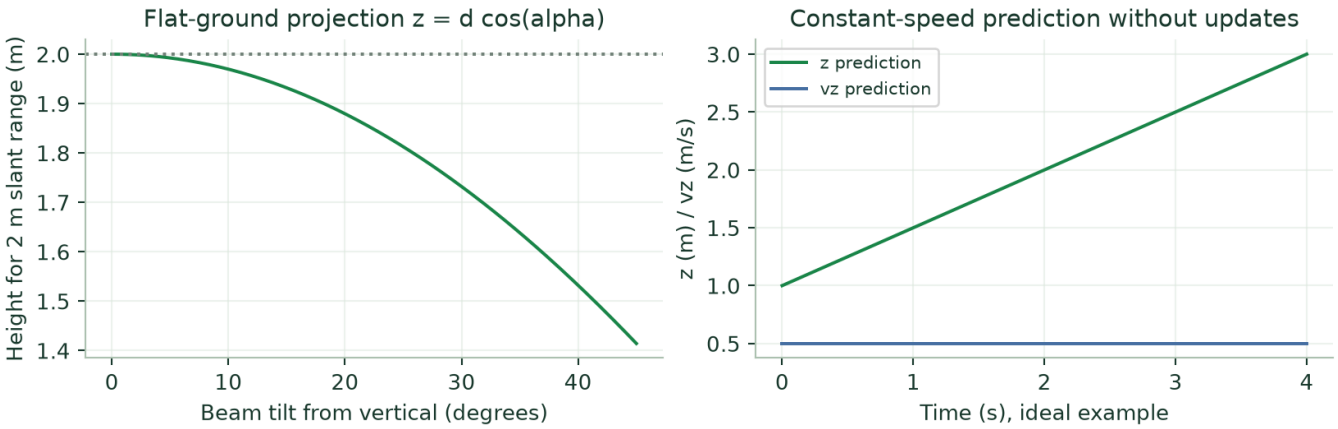
Vertical estimation and sensor geometry



PROPOSED ARCHITECTURE — Two-state estimator and vertical control architecture proposed; barometer and lidar integration remain pending.

$$z_{k+1} = z_k + \Delta t\, v_{z,k} + \tfrac{1}{2}\Delta t^2 a_{z,k}, \qquad v_{z,k+1} = v_{z,k} + \Delta t\, a_{z,k}$$

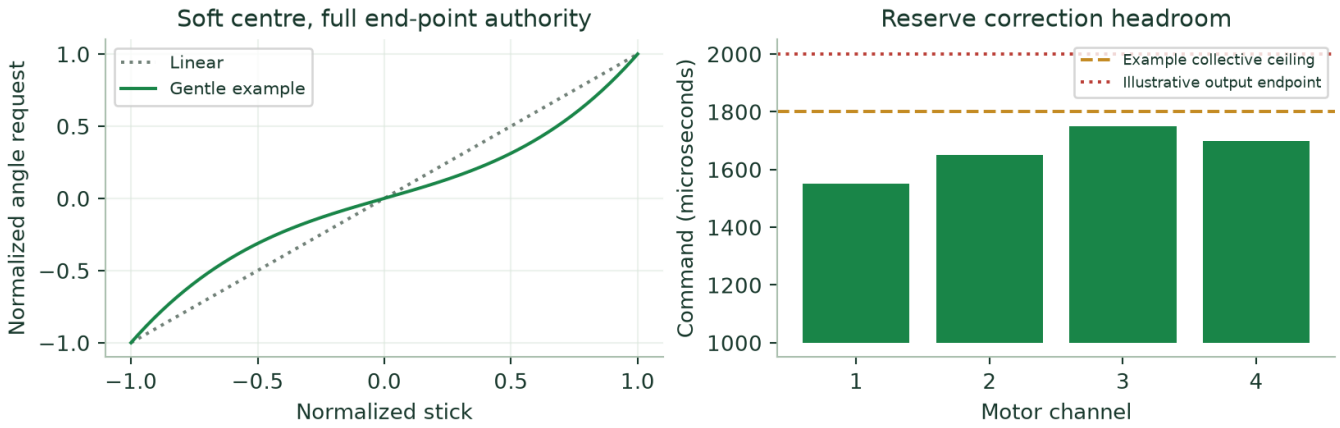
$$P_k^- = A_d P_{k-1} A_d^T + Q_k, \quad K_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1}$$



ANALYTIC EXAMPLE — A forward lidar cannot supply this height observation. Terrain and extrinsics must be checked.

The two-state model estimates z and vz; it does not automatically estimate accelerometer bias. Q must follow the selected disturbance model and sample interval. A mass variance cannot be copied directly into Q. Barometer drift, lidar status, tilt projection, terrain, measurement age and dropout all require tests. Bottom-versus-front lidar placement remains a decision, not an interchangeable software setting.

Gentle control authority and anti windup



ILLUSTRATIVE CONTROL DESIGN — Actual collective/vertical law and ESC limits are TBD. A boost request is not four identical maximum outputs.

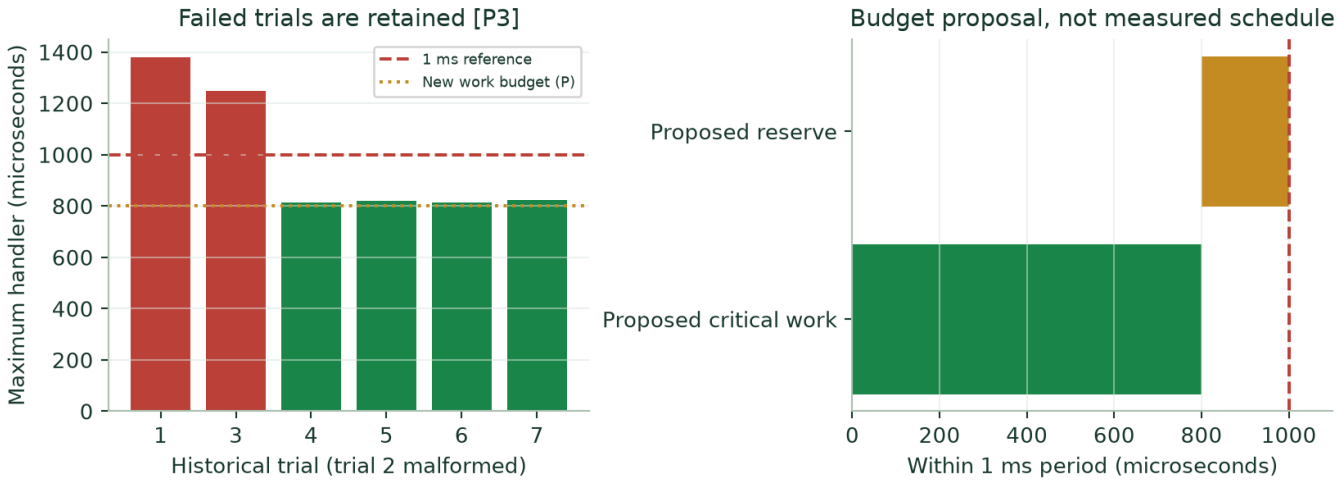
$$u_{raw} = u_P + u_I + u_D, \qquad u = \mathrm{clip}(u_{raw}, u_{min}, u_{max})$$

$$\dot{u}_I = K_i e + K_{aw}(u - u_{raw}) \quad \text{(one antiwindup option)}$$

Design priorities: bounded angle/rate requests, gentle centre response, filtered derivative, explicit saturation, attitude correction headroom and bumpless mode changes. The plotted command values and shaping curve are examples, not firmware settings or accepted ESC endpoints.

Boost is not maximum output. A proposed 1700-1800 microsecond collective ceiling preserves some room for mixing, but the exact relationship between RT/LT, vertical speed and collective remains unresolved. Verify signs, mixer geometry and spin directions before physical actuator work. All motor outputs remain disabled.

Real time evidence including failures



REAL LOG SUMMARY + PROPOSED BUDGET — Historical samples were not a complete PID controller; source hashes and inter-service gaps still matter.

$$T_s = 1000\ \mu s, \qquad T_{critical} + T_{reserve} \leq T_s$$

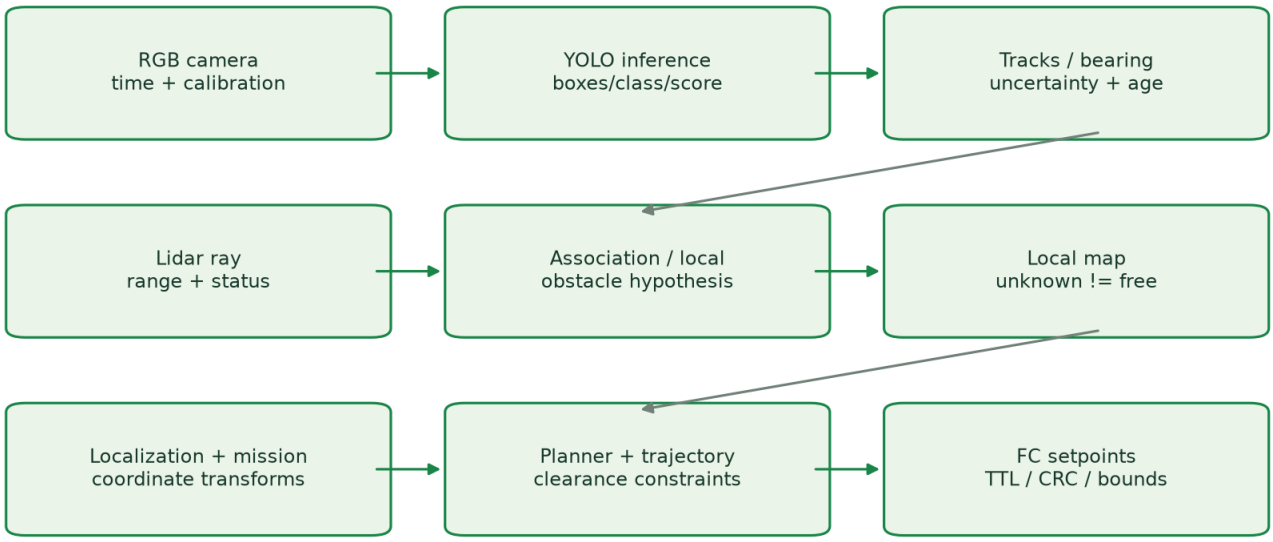
Measured history [P3]: trial 2 was malformed and rejected; earlier trials exceeded the handler budget. Repair separated acquisition/calculation from synchronous serial formatting using a bounded queue. The later 822 microsecond maximum leaves only 178 microseconds to 1 ms if compared directly; a complete future FC includes additional work.

Proposed target: <=800 microseconds critical work, >=200 microseconds reserve, and no observed deadline miss in a defined 10-minute loaded test. Historical samples near 1000.69 Hz do not prove a periodic hard deadline. Inter-service gaps, clocks and source changes remain relevant.

P5 build only: 54 Python checks, 19 actual-source host cases, ESP32 compile; 19504 bytes RAM and 312901 bytes flash. Source f26d314e was not uploaded in that record. Historical timing source 91947b5c is different. A host fake peripheral is not an ESP32 timing measurement.

Bench source configuration: accelerometer +/-8g, gyro +/-500 degrees/s, nominal 260 Hz low-pass setting with readback checks. Sensor bandwidth is not the PID bandwidth. These settings do not validate the revised sensor wiring or a full control loop.

YOLO to obstacle avoidance architecture



PROPOSED ARCHITECTURE — Proposed perception-to-control pipeline. Detection is not depth, localization or a collision-free corridor.

$$\mathbf{z}_{det} = (u_1, v_1, u_2, v_2, c, s), \qquad \mathbf{d}_c \propto K_c^{-1}[u, v, 1]^T$$

$$U^* = \arg\min_U \sum_k \left(w_p \|\mathbf{p}_k - \mathbf{p}_{goal}\|^2 + w_u \|\mathbf{u}_k\|^2 + w_c \psi(d_k - d_{safe}) \right)$$

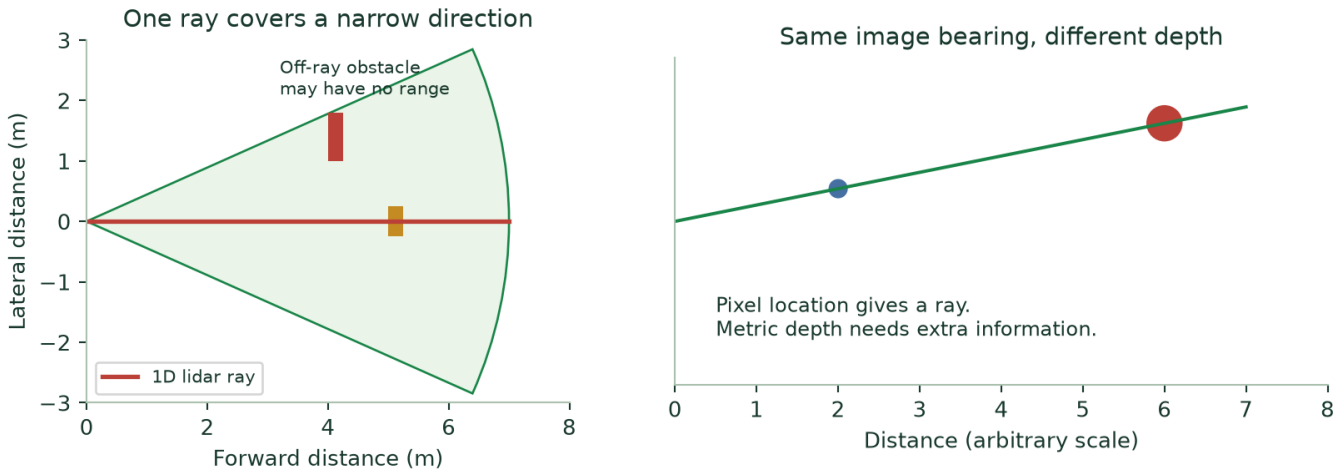
YOLO supplies: bounding boxes, class and score [R7]. It does not supply complete metric free space. Camera intrinsics Kc, camera-to-body extrinsics, timestamps, tracking and uncertainty are additional contracts.

Fusion gate: associate a lidar ray with a visual hypothesis only when geometry, timing and field of view justify it. Unassociated or unobserved space stays unknown. No detection does not imply no obstacle. Localization must make waypoint coordinates observable before a metric mission can execute.

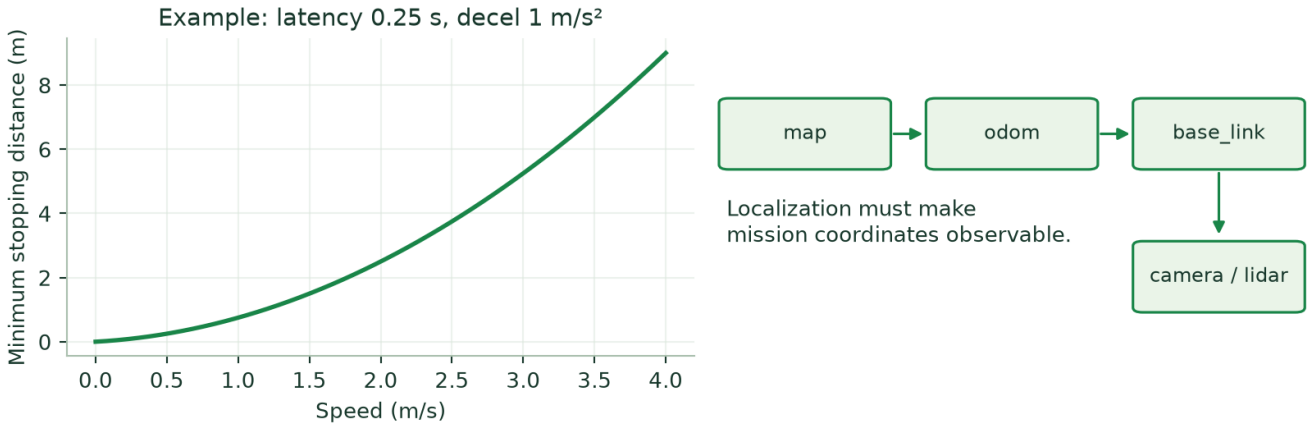
Panel obstacles may be absent from pretrained classes. Training data, panel recognition and a class-agnostic fallback remain to be selected and tested.

Candidate planning objective: penalize goal error, control effort and inadequate clearance, subject to the motion model, input bounds and observed-space constraints. U is a candidate control sequence; psi is a clearance penalty. This does not select or validate a planner. The FC remains responsible for bounded commands. Model weights, runtime, licence, speed and companion hardware remain TBD.

Perception limits and stopping clearance



GEOMETRIC SCHEMATIC — Objects, field of view and distances are illustrative, not camera specifications or YOLO detections.



ANALYTIC EXAMPLE + PROPOSED FRAMES — Real braking, latency and localization are unmeasured; speed limits cannot be inferred from this example.

$$\mathbf{p}_w = R_{wb}\left(R_{b\ell}[d, 0, 0]^T + \mathbf{t}_{b\ell}\right) + \mathbf{t}_{wb}$$

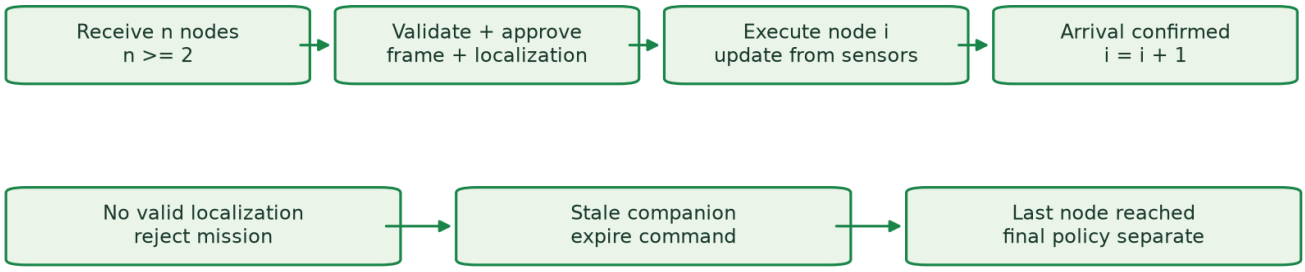
$$d_{clear} > v\tau + \frac{v^2}{2a_{brake}} + d_{margin}$$

One RGB camera plus one fixed ray cannot observe the entire collision corridor. Forward lidar helps one direction; downward lidar helps local height under terrain/tilt assumptions. Metric monocular reconstruction needs additional information and validation; a pixel bearing alone has ambiguous depth.

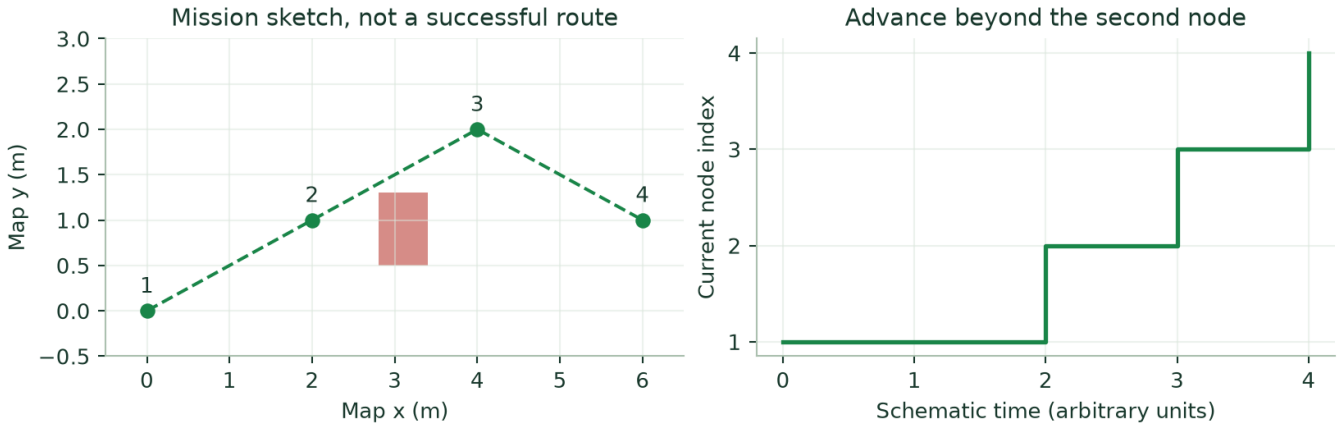
The point transform assumes the lidar ray follows its local positive x axis: rotate/translate lidar to body, then body to world. Extrinsics and pose timestamps must be known; otherwise range cannot be placed reliably in the obstacle map.

The clearance graph assumes 0.25 s total latency and 1 m/s squared braking only to explain scaling. These are not measured aircraft parameters. Reduce the operational envelope until latency, braking, localization and missed-detection behaviour are verified. Any simulator depth camera or perfect pose must be declared as extra sensing or an oracle.

Missions with two three or many nodes



PROPOSED ARCHITECTURE — Mission progression must not stop at node 2. Loss responses and final landing policy require approval.



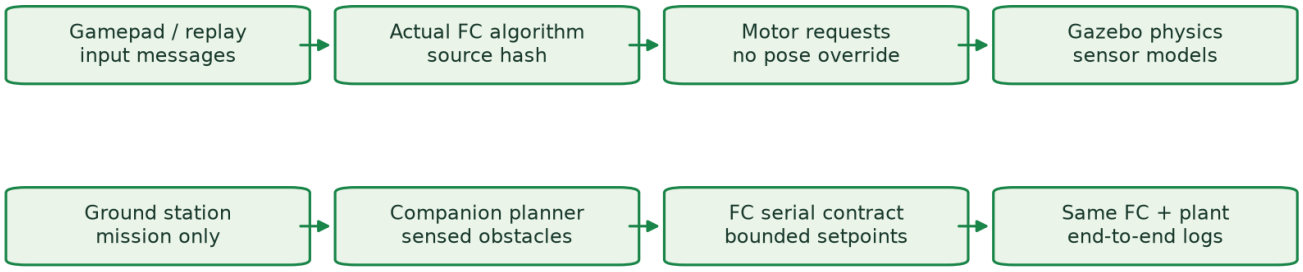
PLANNED TEST SCHEMATIC — The dashed node sequence is input geometry; it does not claim obstacle avoidance or a collision-free path.

$$\|\hat{\mathbf{p}} - \mathbf{p}_i\| \le r_{accept}, \quad \|\hat{\mathbf{v}}\| \le v_{accept} \quad \text{for} \quad t \ge t_{dwell}$$

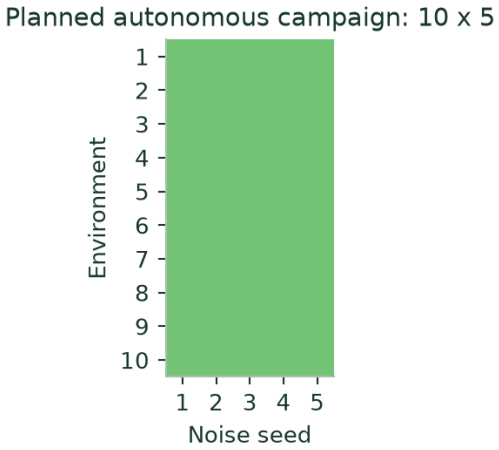
Proposed arrival: radius 0.5 m, dwell 1 s, speed bound TBD; adapt to localization uncertainty. Increment every intermediate node, not only the first. Final arrival and landing/termination are separate policies. The node sketch is an input mission, not a computed safe trajectory.

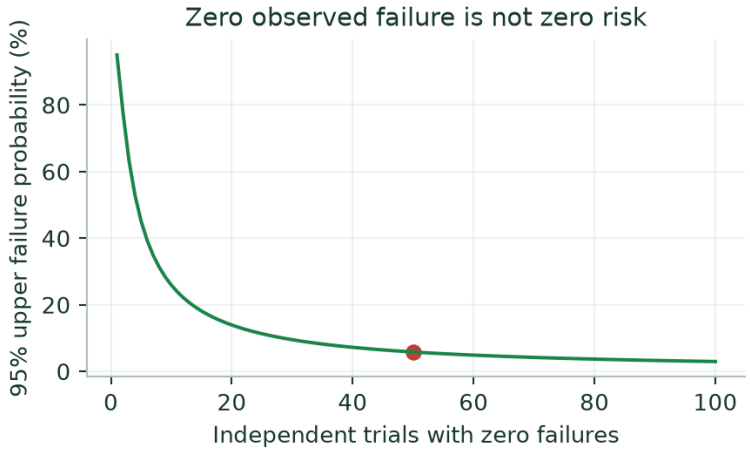
Communication: versioned framed 3.3 V UART is proposed, with length, CRC, sequence, timestamp, units, mission identity and command expiry. Proposed heartbeat 10 Hz and expiry 300 ms require review. Invalid frames do not renew validity. Companions may be absent; plugging one in never changes mode automatically. Loss reactions remain to approve.

Proving that the FC really flies the simulation



PROPOSED ARCHITECTURE — A test harness may stimulate inputs and record outputs; it may not secretly move the aircraft to pass.





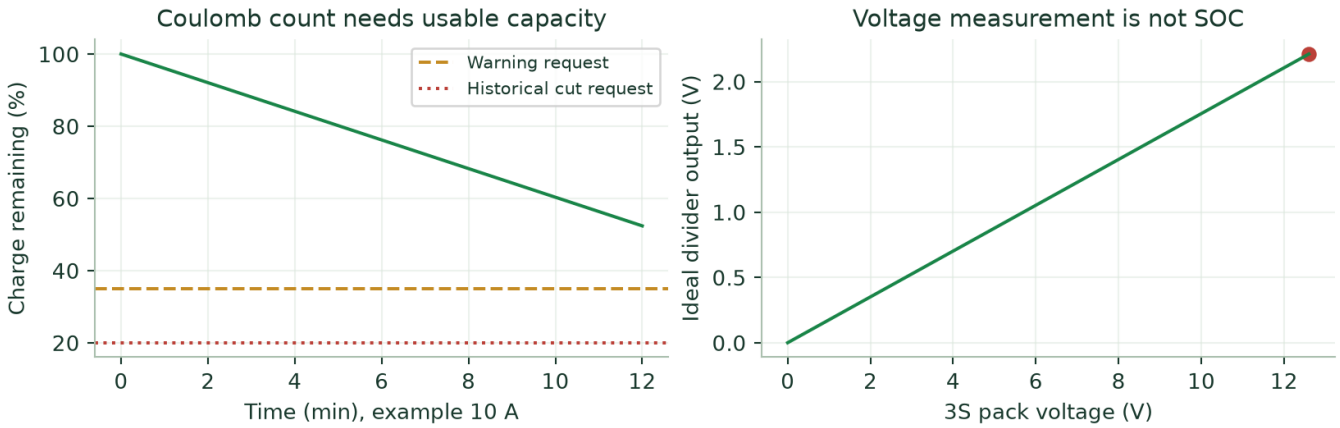
PLANNED CAMPAIGN + STATISTICAL LAW — No cells represent completed missions. Independence and comparable trials are assumptions, not guarantees.

Test contract: stimulate normal input interfaces; log source hashes, messages, estimates and motor requests. Do not apply hidden pose, speed or force corrections to complete a mission. Host-compiled FC code requires disclosed shims; it is not automatically full ESP32 emulation.

Planned Gazebo campaign: ten environments, five noise seeds each; five additional manual command scenarios. Preserve all failures. Video reviewed at 10-15 fps supports trajectory analysis; full-rate logs establish controller timing. No new campaign has been run for this report.

P targets: attitude RMS <=2 degrees, altitude RMS <=0.20 m, endpoint error <=0.5 m, zero collision in the declared campaign. They are reviewable objectives, not achievements. Zero failures in 50 independent comparable trials gives a 95 percent upper failure probability of about 5.8 percent, not zero risk.

Battery estimation and state policy



ANALYTIC EXAMPLE — This is not measured endurance. Whole-pack monitoring cannot protect each cell; a motor cut in flight causes a fall.

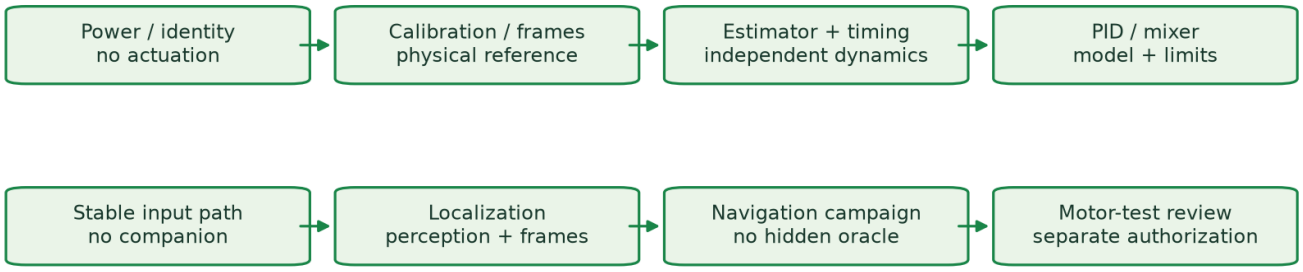
$$SOC(t) = SOC(0) - \frac{1}{C_{usable}} \int_0^t I(\tau)\, d\tau, \qquad V_{ADC} = V_{pack} \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

Measurements: calibrated whole-pack voltage and current, useful capacity, voltage sag and sensor validity. The plots show ideal laws, not measured endurance or an SOC calibration. Marked 120C and four 40 A ESCs do not establish a usable total discharge current.

Historical requested thresholds: warning at 35 percent and motor disarm at 20 percent, without automatic safety landing. A motor cut in flight causes a fall; reconfirm this policy explicitly before any flight phase. Hysteresis, persistence, uncertainty and fault responses remain TBD. Whole-pack software monitoring does not protect individual cells.

Current states: BOOT -> SELF_TEST -> BENCH_READY only. Reset does not arm. Physical motor outputs remain disabled. One verified regulated avionics source; common grounds; no parallel BEC positives; propulsion current stays outside FC copper. Separate authorization is required for motor tests.

Project gates risks and personal development



PROPOSED ARCHITECTURE — Offline preparation is allowed; failed earlier gates invalidate affected downstream acceptance.

Risk or dependency	Engineering response
Wrong axes / signs / motor mapping	Known signed rotations and mixer checks before actuators
Lost data / I2C hang / stale commands	Bounded waits, validity/age checks and explicit degraded state
Saturation / boost / mode transition	Headroom, anti-windup and bumpless references
Missing hardware or deadline	Deliver the verified subset; prepare later stages offline without acceptance
Camera range / localization gaps	Restrict envelope; do not infer unobserved free space

Professional outcomes: requirements -> model -> experiment -> interpretation -> decision. The personal reflection should explain one real difficulty, the student's action, the evidence and the change in understanding. Do not invent experience or sole authorship. Disclose library reuse and AI assistance under institutional policy.

Management: commit each meaningful change; preserve failed evidence; invalidate affected downstream acceptance after an earlier-stage failure. No deadline or rubric was supplied, so the schedule is gate-driven rather than an invented calendar. Add authentic contributions, cover fields, quotations and the actual rubric before submission. Reuse modular hardware and tools; follow appropriate battery, electronic-waste and etching-chemical procedures.

Materials and resources

Installed aircraft BOM: quantities refer to one intended aircraft, not confirmed stock. Verify exact models, physical dimensions, ratings, ownership and masses. Prices are blank until dated quotations are obtained. Optional/future hardware is listed separately on the next page.

ID / qty	Material	Purpose / acceptance
M01 x1	F450 or Q450 X frame with plates and arms	Structural assembly; verify motor diagonal, stiffness and mounting holes
M02 x4	Nominal 1400 KV brushless motors	Exact model TBD; inspect shaft, mounts and propeller compatibility
M03 x4	Nominal 40 A ESCs	Verify voltage range, input protocol and which unit has a suitable BEC
M04 x4	9047 propellers, matched CW and CCW pairs	Verify handedness, hub fit, balance and motor recommendation
M05 x1	3S 4200 mAh LiPo, marked 120C	Inspect condition; weigh; characterize useful capacity and current demand
M06 x1	ESP32-S3 DevKitC-1	Confirm board variant, exposed pins, flash/PSRAM and USB arrangement
M07 x1	GY-521 MPU6050 eight-pin module	Check device identity, pin order, offsets and mounting orientation
M08 x1	BMP280 six-pin module	Confirm BMP rather than BME identity; verify I2C at 0x76
M09 x1	GY-271 style magnetometer module	Identify actual HMC/QMC silicon and matching driver; inspect 3.3 V pull-ups
M10 x1	TOFSense F2 Mini and cable	Verify exact model, saved I2C mode, connector orientation and range status
M11 x1	Xbox Series X/S controller	Confirm firmware, BLE transport, reconnect and input mapping
M12 x1 set	Power distribution assembly	Rated wiring/connectors; common reference; propulsion current outside FC
M13 x1	Suitable regulated avionics supply	Selected verified ESC BEC or separate regulator, not parallel BEC outputs
M14 x1 set	Carrier PCB and module sockets	User-made; fit, continuity, insulation and antenna clearance to verify
M15 x1 set	Fasteners, spacers, battery straps and insulation	Secure battery and rigid sensor reference; prevent conductive contact

Selection logic: ESP32-S3 is the near-term platform; the Pico/radio architecture remains a separate evolution. BLE controller compatibility must be tested with exact controller firmware and library versions [R2,R6]. The magnetometer's actual silicon and driver must be identified rather than trusting the GY-271 board name.

Measurement equipment optional hardware and cost

ID / qty	Material	Purpose / acceptance
M16 x1 set	Battery divider and ADC RC filter	Candidate 47 kOhm / 10 kOhm, 1 percent resistors and 100 nF; verify ADC range
M17 x1	Current measurement assembly	Sensor/range TBD; compatible conditioned output and external high-current path
M18 x4	Three-pin ESC signal headers	Label GND, isolated BEC positive and signal; cable order must be checked
M19 x1 set	Lidar solder pads or connector	Four conductors; strain relief for remote sensor
T01 x1	USB-TTL adapter	3.3 V logic, 921600 baud capability and suitable 5 V lidar power
T02 x1	Digital multimeter	Voltage, resistance and continuity; known uncertainty
T03 x1	Current instrument	Suitable DC clamp or rated inline sensor; never place ammeter across battery
T04 x1 set	Scale, level, angle fixture and ruler	Record resolution and repeatability; rigid fixtures preferred to hand-held poses
T05 x1	3S balance charger and suitable charging setup	Use battery-compatible settings and inspect pack condition
T06 x1 set	Soldering, insulation and inspection tools	Assembly quality, magnification and continuity before power
O01 x1	Raspberry Pi 4/5 or Jetson plus storage	Optional companion; exact model, OS, power and cooling TBD
O02 x1	Compatible RGB camera and mount	Optional perception; calibrate intrinsics, extrinsics and timestamps
O03 x1 set	Companion power and serial interface	Optional; size separately and verify logic level and shared reference
F01 x1 each	RP2350 FC and RP2040 handheld boards	Separate future architecture, not mandatory ESP32 hardware
F02 x2	Compatible LoRa modules	Future main air/handheld link; frequency and local rules to review
F03 x4	CC1101 modules	Future two secondary links per side; not populated for current ESP version

$$C_{base} = \sum_i n_i c_i + C_{shipping} + C_{fabrication}, \qquad C_{planned} = C_{base} + C_{contingency}$$

Use the CSV inventory for owned/to-buy status, vendor, currency, quote date, unit cost and measured mass. Separate shared tools from aircraft cost. A 15 percent contingency with 10/25 percent sensitivity is only a planning scenario. Total cost remains TBD; no supplier price is fabricated.

Verification matrix and proposed acceptance

Test	Coverage	Essential evidence
V01	IF-01, FC-09, COM-01	Power, ground, non-actuating outputs, companion absence
V02	EST-02, IF-02	Cold boot, reset, MPU recovery and readback
V03	EST-01/03/04, MOD-01	Independent poses, signed rotations, uncertainty
V04	IF-03/04, EST-06/07	Known range, invalid returns, removal/restart
V05	RT-01/02/03/04/06	Full-load timing, queue/loss counters and deadlines
V06	FC-02/03/05/07/08, MOD-02/03/04	Model, gains, steps, saturation, source path
V07	HMI-01/02/03/04, FC-01/06	Mapping, gestures, BLE reconnect and expiry
V08	COM-02/03, EST-06, RT-04	CRC, replay, stale/delayed messages
V09	FC-04, EST-05	Vertical response and lost sensor
V10	NAV-01/02/05	All mission nodes, frames and localization
V11	NAV-03/04	Off-ray/unclassified obstacles; no oracle
V12	ENE-01/02/03	Reference power measurement and threshold faults

Every result: commit and source/configuration hashes, engine or hardware identity, original logs, assumptions, fixed criteria, verdict and limits. New gains or configuration invalidate incompatible acceptance. An attractive video is insufficient without the command path and data. No failure is deleted to create a pass.

Assessment evidence: mathematical reasoning, sourced design choices, reproducible original measurements, adverse-result interpretation, resource/risk management and authentic reflection. No guaranteed mark or MIT endorsement is claimed.

Requirements register one

The full engineering baseline is preserved separately. This compact register retains the original requirement identifiers and status, while leaving detailed pinout and API syntax outside the main submission.

ID / status	Requirement	Verify
FC-01 D	Standalone stable mode without companion	V01,V07
FC-02 D	Yaw angular-rate control	V06
FC-03 D	Roll/pitch angle loops with inner rate loops	V06
FC-04 D	Vertical speed input; neutral height hold	V09
FC-05 D	No initial horizontal braking or position hold	V06
FC-06 D	Gentle bounded input shaping	V07
FC-07 P	Boost headroom; collective ceiling 1700-1800 us TBD	V06
FC-08 D	Saturation and integral anti-windup	V06
FC-09 D	Current physical arming and motor outputs disabled	V01
HMI-01 D	Mode 2 sticks; RT increase and LT decrease	V07
HMI-02 D	Down plus A held 5 s requests stable mode	V07
HMI-03 D	Down plus B held 5 s requests ready navigation	V07,V10
HMI-04 D	Down plus LT held 2 s requests disarm	V07
EST-01 D	Calibration tied to mounting, frames and held-out data	V03
EST-02 D	Boot trim restore/readback; no double bias correction	V02
EST-03 D	Gyro bias, scale and physical frame verification	V03
EST-04 D	Angle estimation noise and acceleration limits declared	V03,V06
EST-05 D	Vertical estimator with z and vz states	V09
EST-06 D	Detect stale, repeated, invalid and saturated samples	V04,V08
EST-07 P	Slow observations retain timestamp and age	V04,V05
RT-01 D	Complete FC target 1000 Hz	V05
RT-02 P	Critical-work budget at most 800 us	V05
RT-03 P	No observed miss in defined 10 min loaded test	V05

Requirements register two

ID / status	Requirement	Verify
RT-04 D	Bounded waits, buffers, communication and logging	V05,V08
RT-05 D	One readable Arduino-style main.cpp	Inspect/build
RT-06 D	Distinguish sample, estimation, PID and ESC rates	V05
IF-01 D	3.3 V I2C logic; verified power and common ground	V01
IF-02 D	MPU bypass restore/readback after reset	V02
IF-03 D	Lidar I2C configuration/address verified	V04
IF-04 P	Slow bus starts 100 kHz; changes retested	V04,V05
COM-01 D	Companion optional; no automatic mode change	V01,V07
COM-02 P	Framed/versioned UART, CRC, sequence and validity	V08
COM-03 P	Heartbeat 10 Hz and expiry 300 ms pending review	V08
NAV-01 D	Visit every approved node in order	V10
NAV-02 P	0.5 m radius, 1 s dwell and speed criterion TBD	V10
NAV-03 D	Replan from sensed obstacles; no hidden truth	V11
NAV-04 D	Separate detections, range and observed free space	V11
NAV-05 TBD	Select and validate observable metric localization	V10,V11
MOD-01 D	Explicit coherent world/body/sensor frames	V03
MOD-02 D	Physical/LTI derivations and verified discretization	V06
MOD-03 D	Identified gains, stability margins and saturation tests	V06
MOD-04 D	Separate parameter uncertainty, noise, bias and delays	V06
ENE-01 D	Calibrated whole-pack voltage/current measurement	V12
ENE-02 D	Charge estimate distinguishes capacity, sag and integration	V12
ENE-03 D-review	Historical 35% warning / 20% cut request; reconfirm before flight	V12

Software inventory: PlatformIO/Arduino and Adafruit sensor support; candidate Xbox BLE library; ROS 2 C++; Gazebo physics; RViz visualization; YOLO inference; NAssistant configuration; Python analysis; KiCad; Git/GitHub. Pin versions, licences, model hashes and commands. Isaac Sim is excluded from the current workflow.

Evidence access: original private recordings must be made available to the examiner through an appropriate sanitized evidence package. Public summaries/hashes are not substitutes for the originals. The supplied model parameters are not independently measured aircraft properties.

References and evidence provenance

Primary technical sources: [R1] NASA systems-engineering verification/validation: https://www.nasa.gov/reference/2-0-fundamentals-of-systems-engineering/ . [R2] Espressif ESP32-S3 datasheet and BLE support: https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf . [R3] TDK MPU6050 register map: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Register-Map1.pdf . [R4] Bosch BMP280 datasheet: https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp280-ds001.pdf .

[R5] Nooploop TOFSense-F V2.0, hardware and protocol: https://support.nooploop.com/tofsense/ . [R6] Xbox BLE library maintainer: https://github.com/asukiaaa/arduino-XboxSeriesXControllerESP32 . [R7] Ultralytics detector outputs: https://docs.ultralytics.com/tasks/detect/ . [R8] MIT OCW 16.30 feedback control: https://ocw.mit.edu/courses/16-30-feedback-control-systems-fall-2010/ . [R9] ROS frame conventions: https://www.ros.org/reps/rep-0105.html . Reviewed 28 October 2026; sources support methods, not acceptance of this aircraft.

Project evidence: [P1] measurements/mpu6050/2026-10-25-secured-mount-analysis-v2/report.md and analysis.json; [P2] measurements/mpu6050/2026-10-26-gyro-restart-validation/report.md; [P3] measurements/mpu6050/2026-10-26-fifo-and-gyro-frame-results/report.md; [P4] measurements/mpu6050/2026-10-26-offline-attitude-preparation/report.md and analysis.json; [P5] measurements/software/2026-10-27-input-guard/summary.json. Paths are relative to the Stepwise checkout. Evidence-manifest.json records source hashes and repository links.

Figure provenance: assets/visual/figure-manifest.json identifies every chart as a historical measured summary, replay/model result, analytic example, schematic or proposal. Model-only plots use explicit assumptions and do not represent Gazebo or hardware tests. The F450 artwork is AI-generated using the built-in image tool; image-provenance.json retains its prompt and origin. It depicts a plausible red/white F450, not the student's assembled hardware.

Preserved detail: reference/engineering-baseline-v2.1.md retains the previous detailed engineering discussion, exact interface decisions, original requirements, limitations and project context. This visual edition changes presentation; it neither enables motors nor accepts the estimator, PID or autonomous flight. It does not change firmware or calibration constants.